Science AAAS

Supplementary Materials for

A precessing jet from an active galactic nucleus drives gas outflow from a disk galaxy

Justin A. Kader *et al*.

Corresponding authors: Justin A. Kader, jukader@gmail.com; Vivian U, vivianu@ipac.caltech.edu

Science **391**, 911 (2026)
DOI: 10.1126/science.adp8989

The PDF file includes:

Materials and Methods
Figs. S1 to S8
References

Throughout this work we assume a concordance flat dark-energy cold dark matter cosmological model with a Hubble constant H_0 = 69.3 km s^{-1} Megaparsec^{-1}, dark energy density parameter Ω_Λ = 0.714, and matter density parameter Ω_m = 0.286, based on the 9-year Wilkinson microwave anisotropy probe cosmology (*60*).

JWST observations and analysis

Mid-infrared integral-field spectroscopic observations of VV 340a were conducted with *JWST* MIRI (*61, 62*) in medium resolution spectroscopy (MRS) mode on Universal Time (UT) 2023 March 23-24 as part of general observers cycle 1 program GO-1717. Exposures across the short, medium, and long sub-bands of each of the four wavelength channels resulted in observations that cover the full 4.9 to 28.8 μm range. The fast readout mode with an additional reset between integrations (FASTR1) was chosen to optimize the dynamic range expected in the observations. We used the standard 4-point dither pattern in each sub-band observation to optimize the sampling in the dispersion and cross-dispersion directions. The integration time was 444 s per exposure. Because the source extends beyond the instrument field of view (FOV), we obtained a dedicated background exposure with the same observational configuration in each of the three sub-bands.

The uncalibrated frames were downloaded from the Barbara A. Mikulski archive for space telescopes (MAST) and reduced using the *JWST* science calibration pipeline (*63*) version 1.11.0 in MRS batch mode, which includes steps to apply detector-level corrections, wavelength calibration, flux calibration, background subtraction, and to co-add dither frames. The final calibrated cubes had a FOV of (3.″2×3.″7), (4″.0 ×4.″8), (5.″2×6.″2), and (6.″6×7.″7), and a spatial resolution of 0.″196, 0.″196, 0.″245, and 0.″273 for channels 1, 2, 3 and 4, respectively. We analyzed the collisionally excited forbidden emission line of the Ne^{4+} ion at a rest wavelength ($\lambda_{\rm rest}$)= 14.322 μm, (hereafter [Ne v]), which was observed at a redshifted wavelength of ~14.8 μm in MRS Channel 3. The spectral resolution of the instrument at this wavelength is 6×10^{-3} μm (equivalent to 120 km s^{-1}), corresponding to an instrumental resolving power of ~2500. The same observations included detections of the [Ne v] 24 μm ($\lambda_{\rm rest}$ = 24.318 μm) and [Ne III] 15.6 μm ($\lambda_{\rm rest}$ = 15.555 μm) lines.

We estimated the AGN bolometric luminosity ($L_{\rm AGN,bol}$) using empirical correlations between [Ne v] luminosity and several AGN bolometric indicators (*42*). The flux of the [Ne v] line ($F_{\rm [Ne\,v]}$) was measured in a cylindrical aperture centered on the bright IR nucleus, with a radius set to 1″, which is twice the full width at half maximum (FWHM) of the MIRI point spread function at 14 μm. The flux was $F_{\rm [Ne\,v]} = (1.15 \pm 0.06) \times 10^{-14}$ erg s^{-1} cm^{2} . The statistical uncertainty on the flux was ~5 per cent, derived from the square root of the diagonal of the parameter covariance matrix computed by the Levenberg-Marquardt algorithm used when fitting Gaussian profiles to the emission lines. At the adopted co-moving distance of 157 Mpc, we find $L_{\rm AGN,bol} = (1.28 \pm 0.06) \times 10$ erg s^{-1} = $(3.3 \pm 0.1) \times 10^{10}\ L_\odot$ from the [Ne v] line. Assuming an accretion efficiency of η = 0.1, which is typical for a geometrically thin accretion disk (*43-45*), the black hole accretion rate is $\dot{M}_{\rm Edd} = \eta c^2 L_{\rm AGN,bol} = (2.3 \pm 0.1) \times 10^{-2}\ M_\odot$ yr^{-1}.

Fig. 2 shows velocity channel maps of the [Ne v] coronal emission line observed with MIRI. The systemic velocity channel (Fig. 2A) shows the bright IR nucleus plus collimated nebulae (filaments) directed radially outward on symmetrically opposite sides, to the NE and SW. The coronal line gas to the NE and SW has LOS velocities spanning from systemic to approximately ± 500 km s^{-1}, with the NE side redshifted and the SW side blueshifted. The [Ne v] emission shows two filaments to the SW that form a forked morphology, while in the NE, the [Ne v] lines are double-peaked. The P-V diagram of the [Ne v] line emission was constructed using synthetic two-dimensional slits applied to the MIRI data cube using the software QFitsView (*64*). The continuum emission was sampled in the line-free 12000-4500 km s^{-1} channel and then subtracted from the P–V diagrams. The slit was placed along the filamentary NE–SW structures at PA = 45°, with a central width of 3″ and tapering outward with an opening angle of 3°. The slit position is indicated in Fig. 3A. The resulting P–V diagram is shown in Fig. 3E. Fig. 2 shows velocity channel maps of the [Ne v] 14 μm coronal emission line observed with MIRI. The systemic velocity channel (Fig. 2A) shows the bright IR nucleus plus collimated filaments directed radially outward on symmetrically opposite sides, to the NE and SW. The coronal line gas to the NE and SW has LOS velocities spanning from systemic to ∼+500 km s^{-1} and ∼-500 km s^{-1}, respectively, with the NE side redshifted and the SW side blueshifted. The [Ne v] emission shows two filaments to the SW that form a forked morphology, while in the NE the [Ne v] lines are double-peaked. The P-V diagram of the [Ne v] line emission was constructed using synthetic two-dimensional slits applied to the MIRI data cube using the software QFitsView (*61*). The continuum emission was sampled in the line-free 12000-4500 km s^{-1} channel and then subtracted from the P–V diagrams. The slit was placed along the filamentary NE–SW structures at PA = 45°, with a central width of 3″ and tapering outward with an opening angle of 3°. The slit position is indicated in Fig. 3A. The resulting P–V diagram is shown in Fig. 3E.

Keck observations and data analysis

Optical integral-field spectroscopic observations of VV 340a were obtained with KCWI (*65*) mounted at the right Nasmyth focal station of the Keck II telescope. Observations were made with the instrument configured with the blue high-resolution grating with the small slicer (BH-S mode), and the blue medium and blue low-resolution gratings with the large slicer (BM-L and BL-L modes). The resulting spectral resolving powers were R ~ 4500, 2000, and 900, and the instantaneous spectral coverages were Δλ = 4960 to 5340, 3650 to 4550, and 3500 to 5500 Å, respectively. The BH-S mode had a FOV of 8″ × 20″ with slit width-limited spatial sampling (0.″3 pixel^{-1} sampling rate along the 0.″3-wide slit, i.e., square pixels). The BM-L and BL-L modes had a FOV of 33″ × 20.″4 with slit width-limited spatial sampling (0.″3 pixel^{-1} sampling rate along the 0.″7- and 1.″4-wide slits, respectively). The BM-L and BL-L KCWI observations were obtained on UT 2022 May 23 under clear sky conditions (seeing FWHM ~ 0.″9) as part of Keck observing program 2022A_U213. The BH-S observations were obtained on UT 2023 July 16 under clear skies as part of Keck observing program 2023B_E327. For BM-L and BL-L, we employed a 5-point dithering pattern centered on VV 340a, arranged in a 2 × 2 grid plus a central, overlapping pointing, for continuous coverage across a 60″ × 36″ area, plus one dedicated background sky pointing. For BH-S, there was a single pointing toward the nucleus of VV 340a. Integration times were 1200, 1200, and 600 s, for the BH-S, BM-L and BL-L configurations, respectively, with the same exposure times for science, dark, and background exposures. Basic reduction and wavelength solutions for the spectroscopic data were performed using the data reduction pipeline KCWI_DRP version 1.1 (*66*). We measured an instrumental FWHM of 1, 2 and 4 Å in the BH-S, BM-L, and BL-L configurations, respectively.

Several large cylindrical apertures with radius r = 3″ were placed at the locations shown in Fig. S3A to investigate the ionization mechanisms of the gas. The [O III]/Hβ ratio is used to identify whether gas is ionized by radiation from stars, AGN, or shocks (*67-70*). The disk apertures have log([O III]/Hβ) = 0.25, -0.09, -0.12, at the nucleus, and at positions separated by 10″ (= 6.8 kpc) to the north and 11″ (= 7.5 kpc) south, respectively. Toward the SE nebula and the NW nebula, we find higher line ratios of log([O III]/Hβ) = 0.42 and 0.50, respectively. We find higher line ratios in the two filaments: log([O III]/Hβ) = 1 for the aperture placed 10″ from the nucleus along the NE filament, and log([O III]/Hβ) = 0.84 and 1.03 for two positions along the SW filament, 9″ (=6.1 kpc) and 20″ (= 13.6 kpc) from the nucleus. The SE nebula appears to emerge orthogonally to the disk, and the previously observed Hα arcs *(32)* might be portions of a bubble which has burst, so we posit that this gas is part of a kpc-scale supernova-driven outflow or galactic fountain (*28, 71, 72*). The [O III]/Hβ line ratios measured along the filaments are inconsistent with stellar photoionization, but could be explained by ionization by either a hard radiation field from the AGN or by shock ionization.

P-V diagrams of the [O III] line emission were constructed using the same method as for the [Ne v] line. The continuum emission was sampled in the 1200-1000 km s^{-1} channel and then subtracted from the P–V diagrams. The first slit was placed along the galactic main disk, with a central width of 2″ and tapering outward with an opening angle of 3°. The second slit follows the filamentary NE–SW structures at PA = 38°, with a central width of 3″ and tapering outward with an opening angle of 3°. The slit positions are indicated in Fig. 3A and the P–V diagrams are shown in Fig. 3C-D.

We fitted a model of a composite bulge-disk-halo galactic potential to the [O III] velocity data using the Galpy software (*73, 74*). The bulge was modeled as a power-law density spherical potential with an exponential cut-off, with an inner power-law index of 1.8 and a cut-off radius of 0.1 kpc. The disk was modeled as a Miyamoto-Nagai potential (*75*) with a disk scale length of 5 kpc and a scale height of 5 kpc. The dark matter halo was modeled as a Navarro-Frenk-White potential (*76*) with a scale radius of 2 kpc. The relative normalizations of the bulge, disk, and halo potentials were set to 0.001, 1, and 0.002, respectively, before taking their sum as the model galactic potential. The model disk was projected to an inclination angle of 70° (*24*). The model fitting was performed using Markwardt-Levenberg least-squares minimization (*77*) using the residuals between the [O III] line velocity map and the model LOS velocity field. The model LOS velocities projected along the disk (PA = 0°) and filament (PA = 38°) directions are shown in Fig. 3C-D.

We derived an alternative measurement of $L_{\mathrm{AGN,bol}}$ using the [O III] line flux in a 1″ aperture centered on the nucleus of VV 340a using the KCWI BL-L data. We find an [O III] flux $F_{[\mathrm{OIII}]}$ = $(3 \pm 0.1) \times 10^{-16}$ erg s^{-1} cm^{-2}. Adopting an empirical AGN bolometric correction (*39*), we calculate $L_{\mathrm{AGN,bol}} = (1.1 \pm 0.21) \times 10^{44}$ erg s^{-1} from the [O III] measurement, which is consistent with the value derived from [Ne v] above.

The black hole mass M_{BH} was estimated using the black hole mass-stellar velocity dispersion (M_{BH}–σ_*) scaling relation of nearby AGNs (*48*). Penalized pixel fitting (*78,79*) was used to fit the 3600 to 4400 Å spectral region (containing the Ca II H+K stellar absorption features) with a model stellar line of sight velocity distribution (LOSVD), using a stellar template library (*80*). The best-fitting LOSVD for the central 5″ ×5″ ($r \sim 5$ kpc) indicates an instrumental dispersion-corrected σ_* = 127 ± 16 km s^{-1}. This implies an estimated $M_{\mathrm{BH}} = (3.0 \pm 5.21) \times 10^7\ M_\odot$. From this mass, we calculated the Eddington accretion rate $\dot{M}_{\mathrm{Edd}} = 4\pi G/\eta c\kappa = 6.5^{+11.5}_{-4.16} \times 10^{-1}\ M_\odot \mathrm{yr}^{-1}$, where we assume the standard electron-scattering opacity for fully ionized gas $\kappa = 0.34\ \mathrm{cm}^2\ \mathrm{g}^{-1}$ (*81*). This is about 27×$\dot{M}_{\mathrm{acc}}$, or equivalently $\dot{M}_{\mathrm{acc}} / \dot{M}_{\mathrm{Edd}} = 0.035$. We estimated an Eddington luminosity of $L_{\mathrm{Edd}} = 3.7^{+6.56}_{-2.38} \times 10^{45}$ erg s^{-1} = $1.0^{+1.71}_{-6.23} \times 10^{12}\ L_\odot$, so the Eddington ratio $\lambda_{\mathrm{Edd}} = 3^{+6.0}_{-2.2} \times 10^{-2}$.

VLA data analysis

To investigate the radio continuum emission we used archival VLA C band (4 to 8 GHz) observations in its A and B array configurations (*33*). We re-reduced the archival C band images following previous methods (*82*). The A and B configuration images were combined into a composite image with a 1.′3 × 1.′3 FOV covering VV 340a and part of VV 340S, with a synthesized circular beam FWHM of 0.″4. We also utilized archival VLA L band 1.5 GHz observations taken in the C array configuration to produce a radio continuum image with a 15″ beam (58*, 83*). Both radio continuum images are shown in Fig. S6.

Cylindrical apertures with radii 2.″7 and 2.″0 (Fig. S6) were used to measure the flux of the eastern and western jets respectively, yielding flux densities of 12.4 ± 1.27 mJy and 1.69 ± 0.170 mJy at 1.5 and 6 GHz for the eastern jet, and 8.85 ± 0.940 and 1.14 ± 0.114 mJy for the western jet. These values indicate a spectral index of -1.4 between 1.5 and 6 GHz, which is consistent with synchrotron-dominated emission. The 1.5 GHz flux densities were used to estimate the radio jet kinetic powers, because this image is more sensitive to the diffuse lower-level emission from the jet than the 6 GHz data is. Our calculation used $P_{\rm jet}$- $P_{\rm radio}$ scaling relations calibrated at 1.5 GHz based on observations of X-ray cavities in giant galaxies (*84, 85*). Depending on which scaling relation was used, we find the jet kinetic power was $(0.87 \pm 15.4) \times 10^{43}$ erg s^{-1} for the eastern jet and $(0.68 \pm 11.96) \times 10^{43}$ erg s^{-1} for the western jet (*84*), or $(2.20 \pm 30.31) \times 10^{43}$ erg s^{-1} for the eastern jet and (2.00 ± 27.53) 10^{43} erg s^{-1} for the western jet (*85*). In both cases, there is large (factor of ~10) uncertainty which is dominated by the scatter in the empirically calibrated scaling relations.

The S-shaped morphology of the 6 GHz emission is consistent with predictions from models of precessing radio jets (*37, 38*). Starburst or AGN wind shocks can accelerate relativistic electrons producing synchrotron emission (*86*), however we prefer the precessing jet scenario due to the high degree of symmetry on both sides of the galaxy. We interpret the radio continuum emission as cyclo-synchrotron emission from electrons with initially relativistic speeds that decelerate due to interactions with the interstellar medium (ISM).

To estimate the precession period of the jet, we compared the VLA image with models of ballistic helical precessing jets (*38*). The 3-D jet model parameters include: β, the plasma ejection velocity v normalized to the speed of light c; the inclination angle i of the jet with respect to the LOS; the precession angle ϕ; the precession period P; and the precession axis position angle ψ. We generate a synthetic image of the jet by projecting the jet trace (composed of model jet plasma particles propagating along ballistic trajectories) onto the plane of the sky, construct a 2-D histogram of the plasma particles in each (RA, Dec) cell, and adding Gaussian noise to match the thickness of the jet in the VLA image. We fitted the synthetic image of the model jet to the VLA 6 GHz image using a linear least squares approach to minimize the likelihood function, which is taken to be the collapsed 1-D vector of squared residuals between the observed and synthetic jet images. We fixed the initial plasma velocity to $v = 10^4$ km s^{-1} ($\beta = 0.033$), because this is a typical value for radio jets of this power (*35, 39, 40*). The other parameters were free to vary. The best-fitting model gives a jet precession axis PA of $\psi = 33^{\circ} \pm 23.6^{\circ}$, a precession angle of $\phi = 14^{\circ} \pm 8.8^{\circ}$, an inclination of $i = 71^{\circ} \pm 9.8^{\circ}$, and a precession period of $P = (8.2 \pm 5.48) \times 10^5$ yr. The 6 GHz image and the best-fitting projected 3-D model jet are shown in Fig. 4A-B. The 1-D parameter distributions are shown in Fig. S1.

ALMA observations and analysis

ALMA observations of VV 340a in the CO ($J = 2 \rightarrow 1$) [hereafter CO (2-1)] spectral line and ~1.3 mm continuum were obtained on UT 2023 December 8 (using configuration C-6, 44 antennas with maximum baselines of 2.5 km) and on 2024 April 27 and 28, (configuration C-3, 41 and 43 antennas with maximum baselines of 500 m) as part of project 2023.1.00499.S. The ALMA spectral line setup had four 1.875 GHz (~2400 km s^{-1}) wide spectral windows centered at 223.227 GHz (covering CO (2-1)), 224.922 GHz, 245.606 GHz, and 247.356 GHz, each with a spectral resolution of 1.129 MHz (1.5 km s^{-1}). We imaged the CO (2-1) line emission using the continuum-subtracted calibrated measurement provided by the observatory, which was processed through the ALMA calibration pipeline (*87, 88*). To match the VLA and *JWST* observations, we produced primary beam corrected CO (2-1) line cubes with a beam FWHM of 0.″4, pixel size of 0.″05, and spectral resolution of 6.4 km s^{-1}. We produced moment 0 (integrated intensity), moment 1 (intensity-weighted mean velocity), and moment 2 (velocity dispersion) maps across 133 channels (~840 km s^{-1}). We applied sigma-clipping to each image at 5σ, using a mask from the non-primary beam corrected cube. The resulting moment 0, 1, and 2 maps are shown in Fig. S6.

All the CO emission is concentrated in the disk with no substantial emission seen in the outflow. From the unclipped moment 0 map (root mean square velocity of 0.25 Jy km s^{-1}) we measure the total flux density of the line using a rectangular aperture 4″ wide and 24″ long at PA = 0°, covering the full extent of the molecular disk. The measured CO (2-1) flux density for the disk is 486 ± 49 Jy km s^{-1}, equivalent to $L_{CO(2-1)} = (7.1 \pm 0.7) \times 10^9$ K km s^{-1} pc^2 using previous methods (*89*).

To convert this luminosity to mass we require r_{21}, the ratio of CO 2-1 to CO ($J = 1 \rightarrow 0$) [hereafter CO (1-0)]. We used archival ALMA archival CO (1-0) observations for VV 340 (project code 2017.1.01235.S), reduced in the same way as the CO 2-1 data. We produced a moment 0 map matching the velocity width and range of the CO (2-1) moment map described above (rms 0.24 Jy km s^{-1}). Using the same disk aperture, we estimate a CO (1-0) flux density of 159 ± 9 Jy km s^{-1}. This indicates an integrated $r_{21} = 0.75 \pm 0.09$. For the r_{21} calculation, we lowered the spatial resolution of the CO (2-1) cube to ~0.″8, to match the CO 1-0 observations. Similar r_{21} values have previously been reported local ultra-luminous infrared galaxies ((U)LIRGs), although for VV 340a it is lower than the median (*89*). Using the measured r_{21} ratio and a conversion factor from CO(1-0) to H_2 of 1.8 $M_\odot$ (K km s^{-1} pc^{-2})$^{-1}$ [derived from local (U)LIRGs (*90*)], we derive a total molecular gas mass for the disk of $(1.70 \pm 0.17) \times 10^{10}$ $M_\odot$. Using the 0.″4 CO (2-1) moment 0 map and an aperture size of 1″ × 4″ for each lobe, we derive an upper limit for molecular gas in the outflow of < 1.3 per cent of the molecular mass of the disk.

We use the continuum image at 231.35 GHz (1.3 mm) delivered by the observatory, which has angular resolution 0.″19 × 0.″15, to measure the continuum peak flux density. We measure a peak flux density at the position of the core of 0.24 ± 0.06 mJy $beam^{-1}$, which corresponds to a luminosity of 1.65×10^{39} erg s^{-1}. Assuming this mm flux is coronal emission from the central AGN, we use empirical mm-X-ray correlations for nearby AGN [(*47*), their table 1] to derive $L_{AGN,bol} = 4 \times 10^{44}$ erg s^{-1} from the mm continuum.

Chandra data analysis

Chandra advanced CCD imaging spectrometer (ACIS) observations of VV 340a (*23*) consist of X-ray images in the full (0.4-7 keV), soft (0.4-2 keV) and hard (2-7 keV) bands, and an energy spectrum. The Chandra ACIS soft band (0.5 to 2 keV) X-ray image was smoothed with a Gaussian kernel with $r_{smooth} = 3$ pixels and $\sigma_{smooth} = 1.5$ pixels. The resulting image (Fig. S7) shows X-ray emission aligned and overlapping with the outflow, indicating thermal X-ray emission which we interpret as due to gas shocked by a jet (*86*).

The *Chandra* ACIS hard X-ray (2-10 keV) image was used to derived an alternative AGN bolometric luminosity measurement from. We modeled the spectrum from the central 1″ region as a power law continuum, plus a Gaussian for the broad Fe Kα feature at 6.4 keV. The low quality of the spectrum in this aperture makes it unsuitable for a direct measurement of the column density of the intrinsic absorption toward the AGN, so we adopt a value of the column density $N_{\rm H} = 3.7 \times 10^{23}$ cm^{-2}, typical for local (U)LIRGs at early merger stages (*91*). The resulting intrinsic hard X-ray luminosity is $L_{\rm X} = 1.35^{+3.46}_{-0.97} \times 10^{42}$ erg s^{-1}. Adopting an empirically-derived X-ray bolometric correction for AGNs (*92*), we find $L_{\rm AGN,bol} = 2.1^{+7.54}_{-0.58} \times 10^{43}$ erg s^{-1} from the X-ray data. This value is an order of magnitude smaller than those derived above from the optical, mid-IR, and mm measurements. We therefore regard it as an underestimate; it is more likely that the hard X-rays are attenuated by an obscuring medium with a higher column density than we assumed.

Outflow model

As the radio jet propagates outward from the AGN, it couples with the ISM of the galaxy, driving shocks and entraining gas in an outflow that moves outward radially from the AGN. As the jet precesses, the outflowing gas eventually forms a hollow bicone. At any given moment in time, the jet has a narrow interaction cross section with the ISM. The ISM is clumpy and its density varies strongly as a function of disk scale height. Therefore outflowing gas might not be distributed uniformly on the walls of the bicone.

To test our hypothesis that the structures seen in [Ne v] and [O III] are the incomplete walls of a large-scale biconical outflow, driven by a precessing jet, we compared their kinematics with predictions from an idealized biconical outflow model (*56, 93*). The hollow bicone is defined by the PA and inclination of its axis and the opening angles of its inner and outer walls, $\phi_{\rm in}$ and $\phi_{\rm out}$. De-projected velocities along the bicone walls were modeled using a velocity law of the form $v_r(r \leq r_{\rm t}) = k_1 r$ and $v_r(r > r_{\rm t}) = v_{\rm max} - k_2 r$, where $v_{\rm r}(r)$ is the deprojected radial velocity of the gas (viewed from the AGN), which is a function of radius r; k_1 and k_2 are the rates of linear acceleration and deceleration; and $r_{\rm t}$ is the turnover radius where the maximum velocity $v_{\rm max}$ is reached and deceleration begins.

Predictions from the biconical outflow model were compared with the measured kinematics of the [Ne v] and [O III] gas in two synthetic long slits placed along the directions of maximum elongation of the gas (PA = 45° for [Ne v] and PA = 38° for [O III]), shown in Fig. 3A-B. Fig. 3D and Fig. 3E show the P–V data extracted along these slits for both the observations and the total model in Fig. 3D. The total model is the sum of the velocities from both the outflow model and the galactic dynamical model in each position bin. There is good agreement between the data and model when the bicone has an orientation and inner opening angle set equal to the jet precession axis orientation and precession angle, an outer opening angle of 20°, and a velocity law with $k_1 = 1000$ s^{-1} and $k_2 = 100$ s^{-1}, giving $v_{\rm max} = 1.2 \times 10^3$ km s^{-1} at $r_{\rm t} = 1.25$ kpc.

We calculated the mass outflow rate of gas traced by the [Ne v] 14 μm coronal line using the biconical outflow model $\dot{M}_{\rm out} = 2 m_{\rm p} n_{\rm e} v_{\rm max} A f$ (*56*), where $m_{\rm p}$ is the proton mass, $n_{\rm e}$ is the electron density of the gas, $v_{\rm max}$ is from the bicone model, A is the cross-sectional area of the bicone base (on one side) measured at $r_{\rm t}$, and f is the volume filling factor. The electron density was estimated directly from the [Ne V] 14 μm / 24 μm line flux ratio measured in two 1″ cylindrical apertures, centered on the collimated [Ne V] nebulae to the NE and SW (Fig. 2). We find ratios of 0.97 ± 0.094 and 1.00 ± 0.079, respectively. The theoretical line emissivity software PyNeb (*94, 95*) was employed to determine the electron density from the [Ne v] 14,24 µm line ratio and an estimate of the electron temperature, which we obtained from the [O III] auroral-to-nebular emission line ratio $R_{[\text{O III}]} = 0.02 \pm 0.008$ in the NE aperture, since the [O III] λ4363 auroral line was not detected in the SW. PyNeb indicates that the electron temperature and density in the outflow are $T_e = (1.83 \pm 0.291) \times 10^4$ K and $n_e = (1.07 \pm 0.408) \times 10^3$ cm^{-3}, respectively. The annular cross-sectional area of the bicone at r_t is $A = 2.9 \times 10^5$ pc^2, calculated from the model parameters above, and $v_{max} = 1.2 \times 10^3$ km s^{-1}. Assuming a filling factor of 0.001 (*56*), the ionized gas mass outflow rate $\dot{M}_{out} = 19.4 \pm 7.85\ M$ y^{-1}. In a 1″ -radius cylindrical aperture placed at r_t, the average turbulent velocity in the NE and SW apertures, measured from the broad component of the [Ne v] line, was $\sigma_{turb} = 326 \pm 24.4$ km s^{-1}, leading to an outflow kinetic power $\dot{E}_{out} = ½\dot{M}_{out}(v_{max}^2 + \sigma_{turb}^2) = (1.0 \pm 0.40) \times 10^{43}$ erg s^{-1}. The density measurement was repeated in the NE aperture using the [O II] 3726,3729 Å ratio observed in the BM-L data cube. We find a ratio of $R_{[\text{O II}]} = 1.06 \pm 0.127$, corresponding to an electron density of $n_e = (7.6 \pm 3.10) \times 10^2$ cm^{-3} and $T_e = (1.78 \pm 0.300) \times 10^4$ K. Using this density, the mass outflow rate and outflow kinetic power are $\dot{M}_{out} = 13.8 \pm 5.94\ M_\odot$ y^{-1} and $\dot{E}_{out} = (7.2 \pm 3.55) \times 10^{42}$ erg s^{-1}. All of the outflow rates calculated here are upper limits, because they assume the outflowing gas is uniformly distributed on the bicone surface, contrary to the observations.